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USE OF IP NETWORKS FOR DIGITAL AUDIO TRANSMISSION

BACHELOR THESIS

Study program: Multimedia in the Economic Practice

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Prague, December 2025

Acknowledgement

I would like to express my gratitude to my supervisor, Bc. et Bc. Jaroslav Svoboda MSc for his guidance and assistance. I would also like to mention Filip Němec, who helped me understand what I found incomprehensible.

My deepest thanks go to my family, especially my mum, for her never-ending patience and support.

Abstract

This bachelor's thesis explores the use of IP networks for digital audio transmission, focusing on application in professional broadcast environments. With the ongoing shift from traditional analog and digital point-to-point systems to more flexible and scalable options, audio over IP technologies have gained significant relevance. This thesis presents an overview of digital audio fundamentals, IP networks, and key IP audio solutions such as Dante, AES67, and MADI. The emphasis is placed on synchronization through the Precision Time Protocol (PTP) and the challenges of interoperability between different audio solutions.

The practical section examines the implementation of AoIP in Czech Television's mobile broadcasting unit DSNG46. Using five simplified network models and insights from four semi-structured interviews with engineers and crew members, the thesis analyzes how AoIP technologies influence workflow and production efficiency.

The findings indicate that AoIP enables improved scalability, reduced cabling, and faster reconfiguration, making it suitable for mobile broadcast production. However, it also presents challenges in terms of system complexity, increased demand for network expertise, and potential limitations in device compatibility. Despite these drawbacks, the benefits of AoIP are evident when proper planning, training, and infrastructure are in place.

The thesis concludes that AoIP represents a forward-looking solution in outside broadcasting units and highlights areas for further research and improvement.

Keywords

Audio over IP (AoIP), IP networks, digital audio, broadcast technology, synchronization, AES67, Dante, MADI, interoperability, DSNG

Abstrakt

Tato bakalářská práce se zabývá využitím IP sítí pro přenos digitálního zvuku se zaměřením na jejich aplikaci v profesionálním televizním vysílání. S rostoucím přechodem od tradičních analogových a bodových digitálních systémů k flexibilnějším a škálovatelnějším řešením založeným na sítích nabývá technologie Audio over IP (AoIP) na významu. Práce představuje přehled základů digitálního zvuku, architektury IP sítí a klíčových technologických řešení, jako jsou Dante, AES67 a MADI. Zvláštní důraz je kladen na synchronizaci pomocí protokolu Precision Time Protocol (PTP) a výzvy spojené s interoperabilitou různých systémů.

Praktická část zkoumá zavedení AoIP v mobilní vysílací jednotce DSNG46 České televize. Pomocí pěti zjednodušených modelů sítě a čtyř polo-strukturovaných rozhovorů s inženýry a členy štábu práce analyzuje, jak tyto technologie ovlivnily pracovní postupy a efektivitu výroby.

Zjištění ukazují, že AoIP umožňuje lepší škálovatelnost, méně kabeláže a rychlejší rekonfiguraci systému, což je výhodné zejména pro mobilní vysílání. Na druhou stranu však technologie přináší vyšší nároky na odborné znalosti, zejména v rámci sítí, složitější správu a možné omezení v kompatibilitě zařízení. Přes tyto výzvy se ukazuje, že při správném plánování a zaškolení má AoIP výrazný přínos.

Práce dospívá k závěru, že AoIP představuje perspektivní řešení pro moderní potřeby televizní produkce a poukazuje na oblasti vhodné pro další výzkum a zlepšení.

Klíčová slova

Audio over IP (AoIP), IP sítě, digitální zvuk, vysílací technologie, synchronizace, AES67, Dante, MADI, interoperabilita, DSNG

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Introduction

The evolution of audio technology has significantly influenced the way media are produced, transmitted, and consumed. Over the past two decades, the broadcasting industry has been undergoing a steady shift from analog and traditional digital formats toward more flexible, scalable, and IT-integrated solutions. One of the most significant advances in this area is the use of IP (Internet Protocol) networks for digital audio transmission, commonly referred to as Audio over IP (AoIP). These systems offer a wide range of advantages, including reduced cabling, improved routing flexibility, and better integration with existing IT infrastructures.

This thesis focuses on the use of IP networks for digital audio transmission, with a particular emphasis on their practical implementation in the broadcasting environment. The topic was chosen based on a growing need within broadcast operations to modernize infrastructure, improve efficiency, and adapt to hybrid production workflows that combine on-site and remote elements. The inspiration for this work also comes from the author's direct involvement in the redesign and upgrading of the DSNG46 (Digital Satellite News Gathering) unit at Czech Television.

The main objective of this thesis is to examine the benefits and challenges associated with the deployment of AoIP systems in professional broadcasting. The work seeks to answer whether AoIP improves production quality, simplifies workflows, and enhances flexibility in a real-world mobile broadcast setting.

Methodologically, the thesis combines a theoretical and applied approach. The theoretical part introduces key concepts of digital audio, IP networking, and AoIP standards. The practical part focuses on the DSNG46 unit at Czech Television, where the author participated in the implementation of a new audio infrastructure. This section includes three simplified technical models and a series of semi-structured interviews with system engineers and operators to evaluate the real-world impact of the AoIP integration.

This work aims to provide valuable information on the current capabilities of AoIP, its practical implications in broadcasting, and considerations for future improvements and implementations.

1 Sound

To comprehend different types of audio representation and their comparisons, one must first understand what the sound is. It has several forms in which it can be perceived, such as physical, analog, digital, and graphical. In this first chapter, the basic principles of audio will be explained, as well as the difference between analog and digital audio.

1.1 Sound as a physical phenomenon

Sound is created by a physical vibration that moves through medium. It is perceived by our bodies, more specifically our ears. D’Escriván (2012) explains how a metal sheet stuck by a rock creates changes in air pressure. Compressions and rarefactions of the air molecules take place as they move through the medium, and this phenomenon can be perceived by our ears or, for example, the membrane of a microphone.

These changes in air pressure can be characterized by their strength and speed. The strength at every exact moment it is measured in is called amplitude, and the speed of the changes is called frequency. Furthermore, as these changes occur periodically, we also measure how many times these periods occur per second, and the unit used is Hertz.

1.2 Analog audio signal and recording

The first successful attempt to capture sound and create a recording of it was in 1857, resulting in a machine known as a phonautograph. This device produced recordings in the form of phonautograms. It gave the basis for phonograph cylinders to be invented later in 1877 by Thomas A. Edison (Friedland & Ramesh Jain, 2014). A cylinder covered in tinfoil was spun around, and a needle placed onto it read mechanical engravings that were recorded on the surface, turning them into sound waves to be amplified.

To capture the waves of vibrations that ripple through air into electrical current, special membranes have been created that can simulate the way our eardrum functions. They reform the changes in air pressure into mechanical vibrations that are then turned into electrical current by devices called transducers. These devices capture the strength of the vibration of the membrane in the form of electrical impulses (d’Escriván, 2012). These impulses are still a part of the same electrical current, making them a continuous signal.

In the twentieth century, the two main methods of recording were mechanical on a vinyl and electromagnetic on a tape. Vinyl used grooves and tape magnetized areas to represent electrical current. These methods were inspired by devices mentioned previously that recorded sounds as engravings of a needle.

1.3 Digital audio

Television production and broadcast have used digital audio instead of analog since the early 2000s. This has changed the way content is recorded, produced, and distributed. Although digital audio has not entirely replaced analog audio, progress has been underway for a considerable period of time.

1.3.1 What is digital audio

Digital audio is a type of representation of an audio signal, commonly encoded as numerical sequences of samples. Thanks to its digital format, computers and other digital devices can store, edit, and produce audio. Unlike analog, a continuous wave of an electric signal, digital audio must be transformed into an array of numbers (Watkinson, 2013).

The sound waves that are recorded are characterized by amplitude, which defines the perceived loudness of the sound, frequency, which interprets the pitch, and waveform shape. These factors can be represented by a binary string of ones and zeros for computers to process and edit (Fischer, 2010).

1.3.2 Conversion

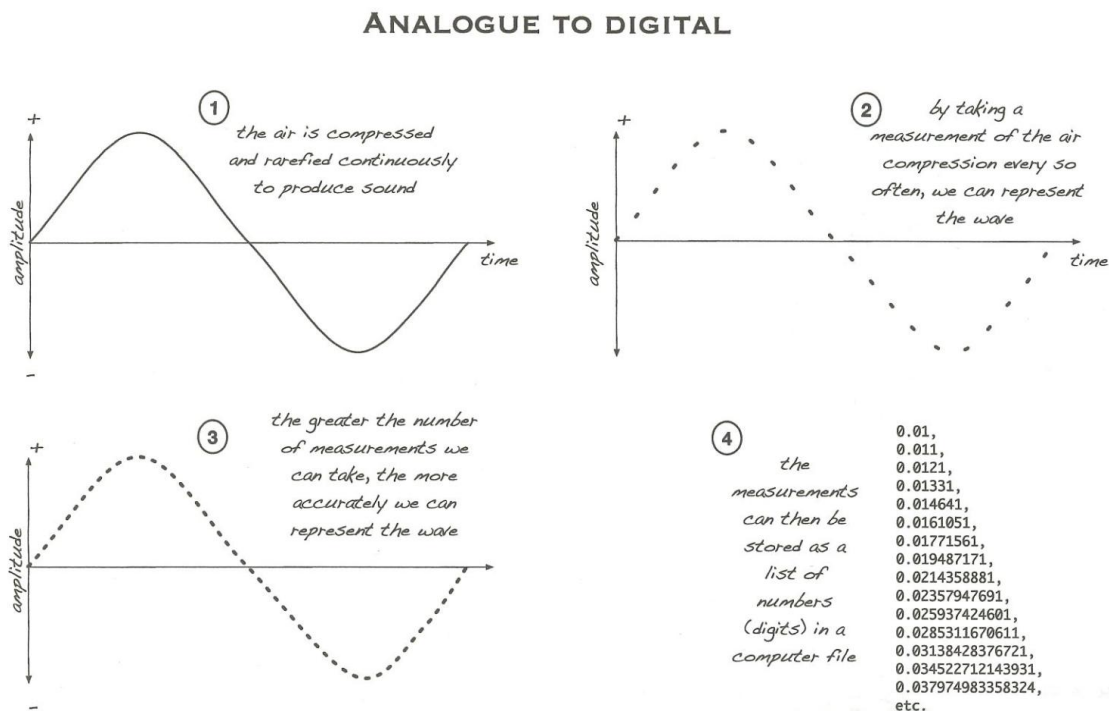
To process audio on computers and other devices, we first must turn the analog signal into a digital one. The device that does this is called an analog-to-digital converter, as it transforms sound into numbers. To turn digital audio back into an analog signal, a digital-to-analog converter is used. It is a device that changes the numbers back into electrical current (d'Escriván, 2012). The process of analog-to-digital conversion turns the continuous signal into a discrete one that is made up of a binary string consisting of 1s and 0s. This process consists of three main steps: sampling - splitting the signal into a list of values; quantization - measuring the amplitudes; and Pulse Code Modulation (PCM) - representing the resulting values in binary numbers.

1.3.2.1 *Sampling*

The sound wave must be broken down into small segments in this step. Each measured point can then have its value. The smaller the segments or measurements, the closer the result is to the original, as seen in Figure 1 on the x axis. These points are called samples and are measured at a sample rate, number of samples per second, in frequency units Hertz (Hz) (Zölzer, 2022).

Figure 1

Analog to Digital Conversion



Note Reprinted from Music Technology (p. 8), by Julio d'Escriván, 2012, Cambridge University Press. Copyright 2012 by Julio d'Escriván

The most commonly used sampling rates are shown in Table 1. In agreement with the Nyquist-Shannon theorem, the sampling rate must be at least twice the signal's highest frequency to reconstruct a signal precisely, as described by Shannon (1949). As human hearing ranges from 12 to 20 kHz, the most frequently used sampling rates are all above 40kHz, covering the entire spectrum of human hearing without loss of samples or distortion (d'Escriván, 2012). For simple recordings, such as only human voice, even lower sampling frequencies may be suitable, covering the range of human voice while keeping the file size low.

Table 1

Commonly used audio sampling frequencies

Sampling frequency in kHz	Common use
22.05	Web Audio API, older Voice-over-IP
44.1	CD recordings, music
48	Digital video production such as movies, TV production
96 and 192	High-Definition audio for digital audio workstations

Note Adapted from *Music Technology* (p. 10), by Julio d’Escriván, 2012, Cambridge University Press. Copyright 2012 by Julio d’Escriván

1.3.2.2 Quantization

Quantization is used to add values to individual samples. It refers to the number of bits used to represent the amplitude or bit depth of a sample (Zölzer, 2022). The higher the bit depth, the more detail and dynamic range the reconstructed signal has.

The typical bit depths for digital audio are 16-bit, 24-bit, and 32-bit. “Audio signals are usually sampled with 16 bits or more. 16-bit resolution, however, still does not match the dynamic range of the human ear, which is about 140 dB” (Fischer, 2010, p. 156).

1.3.3 Formats

To store and transmit digital audio, we need to save it in an audio format. These formats can be split into two categories, uncompressed and compressed. The compressed formats are then divided by the type of compression into lossless and lossy formats.

Uncompressed formats store all the data exactly as they were captured, without losing any information. However, this means that the resulting file size is larger compared to lossy formats, such as MP3. The most commonly used uncompressed formats are .WAV and .AIFF (Griffey, 2024). WAV was developed by Microsoft, so it is used on all Windows-based platforms. AIFF is an Apple alternative to WAV. These formats are widely used for professional work with audio and post-production.

Lossless formats compress the file without losing any quality. The file size is smaller than that of an uncompressed file, and the original can be retrieved from the losslessly compressed file. The most widely used lossless file formats are FLAC and ALAC (Hans & Schafer, 2001).

Lossy compression means that some parts of the recorded information are lost in the process. On the other hand, it produces the smallest file sizes. The user can define the strength of the compression, determining the amount of discarded quality and the file size. These formats are

standardly used for distribution amongst the public (Griffey, 2024). Formats such as MP3, Opus, and AAC belong to this category.

1.3.4 Digital vs Analog

While digital audio has been on the rise since the early 1980s, analog audio still has its uses and advantages and is still being used in today's production. From a technical point of view, there are several differences between these two types of audio.

One of the leading advantages of digital audio is its nondestructive ways to edit it. Its files can be easily edited, copied, cut, filtered, and processed using many available software products on the market. This can be done with the ability to undo any mistakes and in a user-friendly environment. In comparison, editing analog audio is usually destructive and physical (Savage, 2011). For example, something cannot be reversed once something is cut from magnetic tape.

Another significant difference is in the way audio is reproduced and distributed. To distribute analog sound in media such as tapes or vinyl, a physical copy must be made. These can be expensive to create as well as distribute to customers. Physical media such as vinyl also degrade with time and each use. On the other hand, digital media can be instantly distributed across the world using the Internet without loss of quality.

Analog audio also has high inherent noise. "Noise can be defined as any unwanted sound that is not related to the wanted sound (if it is related, it is called distortion)" (White & Louie, 2005, p. 259). Tape recorders that have magnetic characteristics also suffer from modulation noise, which adds more unwanted disturbances. Although digital audio systems also suffer from inherent noise, their noise floor is much lower. Some modulation noise can be added by using analog-to-digital converters.

If the sampling frequency in a digital audio system is set too low, aliasing can occur. This is a mistake in the conversion of the signal to digital sound that results in distorted sound. The aliasing cannot be fixed once it has been recorded.

Inaccuracies that only analog audio media suffer from are flutter and wow. It is caused by irregular movement of the tape or recording, and the result is a change in the pitch of the sound (White & Louie, 2005).

Although analog audio has more disadvantages from a technical point of view, some audiophiles prefer it to digital form. This disposition often stems from nostalgia and the perceived sonic 'warmth' or characteristic sound signature of older playback equipment. Although digital audio has replaced analog technology in numerous domains, certain devices and technologies, for example analog microphones, retain an essential role and remain difficult to substitute.

2 IP Networks

Computer networks have been connecting the world for several decades now. It helps people connect worldwide, send and receive files, and collaborate regardless of their location. This has changed the possibilities of production of television broadcasts and news delivery to the public. This chapter explains the basic principles of IP networking and how it can make collaboration even over state lines easier.

2.1 Fundamentals

An elemental building block of networking is the Internet Protocol (IP). That is an internet protocol as defined previously by Postel (1981). Protocols are a set of predetermined policies that determine the routing and formatting of the sent data. A more specific networking protocol defines the communication between two parts of software or hardware. These predefined rules determine what the message and its encoding should look like and what information should be added to the conversation.

“A protocol defines the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on the transmission and/or receipt of a message or other event” (Kurose & Ross, 2017, p. 8).

Information and messages that are exchanged between two or more entities are split into smaller chunks called packets. They are prepared and transmitted by the sender, and on their way to the receiver, they usually travel through packet switches. Packets obtain not only the message that a device is trying to send, but also protocols that add information to it. This information is stored in an IP header, containing information about essential routing and delivery information, such as source and destination IP addresses, packet length, Time-to-Live (TTL) value for IPv4 or Hop Limit for IPv6, and protocol type (Misra & Goswami, 2017). The header also includes key information, such as the IP version and the header length.

An IP network is a collection of interconnected devices that can communicate with each other. These networks can operate over local area networks (LANs), wide area networks (WANs), or the global Internet. Each connected device has its own IP address so that devices in the network to which it is connected can locate them. When a packet is sent, it will contain the location of both the sender and the recipient (Kurose & Ross, 2017).

2.2 TCP and UDP in Digital Audio Transmission

Transport protocols must be put in place to send digital audio or any other information over the network. They ensure the reliability and speed of the transmission. The two most commonly used transport protocols are the Transmission Control Protocol (TCP) and User Datagram Protocol (UDP). Both try to provide logical communication between entities. This means providing connectivity as if the devices were immediately connected to each other, although they can be hundreds of kilometers away with several devices in between (Kurose & Ross, 2017). Each transport protocol serves different purposes depending on the nature of the transmission and its requirements for reliability, latency, and bandwidth.

TCP focuses on reliable connections and data delivery. It provides a connection-oriented service. TCP ensures that before any data transfer starts, both entities communicate and are ready for bigger data transport. This is called a handshake between a client and a server. TCP adds a transport header in front of each segment, which helps to ensure that all packets are delivered and are in the correct order, as stated by Kurose & Ross (2017). When a device sends a segment, it needs a response from the recipient that it has received the whole packet and that it is in the proper order. TCP will request redelivery if any packet is lost or corrupted in the transport process or if a receipt is not delivered (Leung et al., 2007). This assurance makes the Transmission Control Protocol ideal for the safe delivery of segments, but it suffers due to speed. It is better for non-real-time audio applications such as file transfers, on-demand streaming, or configuration data exchange.

In contrast, UDP focuses on speed but loses reliability. Kurose & Ross (2017) characterize it as a connectionless protocol that sends the datagram without establishing a connection or a handshake and does not require receipts. It also does not verify whether the packets arrived in the correct order. As UDP provides very low latency, it is suitable for real-time audio broadcasts, such as streaming, live news broadcasts, and conferencing. Although UDP does not correct for packet loss, professional audio systems typically implement higher-layer mechanisms to handle these challenges, such as buffering, forward error correction, or redundant streaming (Xu & Cooperstock, 2000).

2.3 Transmission Models: Broadcast, Unicast, and Multicast

Another significant aspect on which networks are relying is the communication model. There are three primary models: unicast, broadcast, and multicast. Each method is distinguished based on how many receivers there are at the end and serves their specific purpose in distributing digital audio.

2.3.1 Unicast

Unicast establishes a one-to-one connection and is optimally used when data is intended for a single recipient. In this communication model, the sender uses the destination IP address for global routing to the correct network, while the MAC address is only used for local delivery within the final network segment (Misra & Goswami, 2017). However, unicast communication is highly inefficient when multiple recipients request the same data simultaneously. Generating individual data streams for each recipient leads to a large volume of replicated traffic, which can result in significant network congestion.

2.3.2 Broadcast

Broadcast communicates with all devices on the available network, regardless of whether the recipient requested or wanted the data; it is a one-to-all method in which one sender transmits a message to every device connected to the same local network. Every device in the network segment receives this message and must process it (Misra & Goswami, 2017).

This method is usually used to send out protocols when all devices need data. However, because every device must stop and listen, broadcast is highly inefficient for sending large amounts of application data. If too much is broadcast, the network becomes congested and slows down for everyone.

2.3.3 Multicast

In contrast to the above-mentioned options, multicast offers a distribution method that is applicable in digital audio transmission for professional networks. It offers one-to-many transport of data, as well as offering the possibility of several senders and receivers. In this method, the primary sender sends one data stream to a specific group that signed up to receive it. It can also branch out and let the receivers be senders for different groups of receivers. This approach considers the bandwidth used by the network by allowing it to replicate and send packets only at points where the stream branches out to multiple recipients (Misra & Goswami, 2017).

Multicast uses the Internet Group Management Protocol, IGMP. It is a protocol that helps routers find only the devices that want to be subscribed to the multicast stream and form a group of them. There have been two previous versions; the actual version used is IGMP v3 (Cain et al., 2002). Without this protocol, multicast would behave like a broadcast and would try to send packets to all connected devices, causing flooding of the network.

To allow devices to subscribe to certain multicast streams and others to not receive it, IGMP snooping is used. These devices would receive the stream and then decide whether to accept or deny the stream. When IGMP snooping is allowed on the switch, it can discover which devices are subscribed to the stream and send it to only those, bypassing those that do not want to receive it (Vidal, 2019).

For these reasons, multicast is most often used in systems that distribute high-quality and low-latency audio to multiple endpoints, such as in broadcast studios, large-scale live recordings, and multi-room audio networks.

To conclude this chapter, every type of communication model has its uses, although some are less preferable when dealing with digital audio transmission. The choice between these transmission models significantly affects the performance and scalability of audio over IP systems. Unicast is suitable for direct communications, such as audio conferencing or remote voice-over recording, while multicast is preferred in scenarios requiring simultaneous distribution to many devices. Due to its inefficiency, broadcast is rarely used for digital audio production.

3 Digital Audio Transmission over IP (AoIP)

Every production company tries to find the best compromise between quality, schedule, and price. Usually, the harsh reality is that they can only pick two at the loss of the third. Audio over IP leveraged different fields to bring these three terms of production closer together and make them achievable. Unlike traditional analog or point-to-point digital systems, it enables the delivery of high-quality audio through existing network infrastructure, offering benefits such as scalability, flexibility, and integration with present IT systems.

3.1 Fundamentals

Digital audio transmission over IP networks involves several steps that have already been mentioned in previous chapters. Analog audio signals must be converted to digital data and then segmented into packets, which are subsequently transported through a network to one or more receivers (Church & Pizzi, 2010). The processing software and various protocols add significant information to the packets, also known as metadata. It is hidden data that contain information such as location, creator, timing report, and routing. Devices such as audio interfaces, digital signal processors, and network switches send, route, and receive these packets in real-time. Using IP networks to deliver audio enables simple assimilation of other digital facilities (Church & Pizzi, 2010).

The shift to digital audio networking also contributed to the change in the media used for transport. Unlike traditional analog audio which requires one cable per channel, and point-to-point digital connections like AES3 and S/PDIF, which offer limited channel capacity and fixed hardware routing, AoIP adopted the usage of standardized IP networks. Although AoIP primarily uses twisted-pair Ethernet cables due to them being common and cost-effective, it can also seamlessly utilize fiber optic cables for long distances and high bandwidth, or, less commonly, coaxial cable. In AoIP systems, a single network cable, such as a CAT6 Ethernet cable or a fiber link, can carry multiple hundreds of audio channels, simplifying infrastructure and allowing signals to be dynamically routed via software (Bouillot et al., 2009).

3.2 Synchronization and Timing

One of the most crucial challenges in Audio over IP systems is ensuring harmonious synchronization and accurate timing between interconnected devices. Clock synchronization is essential to prevent audio artifacts, such as clicks, dropouts, or phase issues, that can significantly disrupt the broadcast flow to the audience. The topology of the synchronization networks consists of the Grandmaster clock, one device with the correct time, a time transmitter, and time receivers, those who follow (IEEE, 2019).

For decades now, as Ferrant et al. (2013) depict, global navigation satellite systems (GNSS), such as GPS, have been used frequently to synchronize devices, although they are unreliable in areas where the signal is hard to reach. GNSS uses a system of satellites around the globe. They are all synchronized with precise time. Receivers can get time information from four or more satellites and calculate their own time deviation (Ferrant et al., 2013). If the antenna position of the device is already known, the connection to only one satellite is needed to receive accurate

time using the same coordinate system. GNSS offers a reliable and readily available source of time information, enabling precise time synchronization across various systems and applications without the need for physical connections or expensive atomic clocks.

Many AoIP systems implemented the Precision Time Protocol (PTP) to improve this situation, as defined in IEEE 1588. The Institute of Electrical and Electronics Engineers (IEEE) created this standard for network timing because it can achieve submicrosecond accuracy. This makes it a perfect candidate to be used for television and hence video and audio broadcasts.

“PTP works in master-slave topology. One server or customized device is nominated as the master clock, and all other devices within the subnet synchronize to it forming a network of synchronized servers” (Orme, 2023, p. 31). To provide the best quality of service (QoS), network engineers must set up routers that permit PTP timing packets to ensure rapid and seamless data transmission. A significant part of the network setup is the master clock. It is usually connected to GPS or has a rigorous internal oscillator to provide accurate 1GHz clock samples. This clock is used by PTP so that it can deliver the accuracy of the receiver clocks approximately 1ns (Orme, 2023).

3.3 Interoperability

A critical consideration in creating and utilizing Audio over IP networks is interoperability, the ability of equipment from different manufacturers to communicate and function seamlessly within a shared network. In an ideal world, broadcasters could select any device or piece of technology they prefer, add it to their network of interconnected devices, and start using it among other entities. Unfortunately, that is rarely the case (Turchet et al., 2020).

In early implementations of AoIP, proprietary protocols often created isolated ecosystems where components from one vendor could not integrate with others. This limited flexibility, increased costs, and hindered the expansion of the system. This brand-centered intra-operability reduces the choice that customers and broadcasters have when choosing their devices (Audinate, 2015). This is extremely limiting to users, as they choose equipment for reasons beyond connectivity or staying true to one brand. “Routers and mixing consoles of one manufacturer will not work as plug n play with those of another manufacturer in the same operation as the industry had always been able to do in analog” (McGinley, 2017, p. 1).

The optimal solution for designers, installers, and users would be a fully inter-operable system. This means an ecosystem of devices that communicate with each other, deliver, process, and broadcast audio without any new needed support. It is possible to have a system that they can easily understand, maintain, and use. They desire to have a well-supported, trusted and known solution regardless of the mix of vendors.

A golden middle way would be partial interoperability, where devices from multiple vendors are carefully integrated into a network and meticulously worked around to achieve a seamless workplace (Audinate, 2015). This requires a significant amount of expertise in these topics and can be challenging to maintain.

Today, interoperability has improved because of growing industry support for standardized transport protocols and media formats. Initiatives such as the AES67 standard, which will be discussed in a later chapter, provide a common framework for basic interoperability between otherwise incompatible AoIP systems. It defines parameters for audio format, synchronization, and transport over IP networks, enabling systems based on different protocols to exchange audio streams (Orme, 2023).

With remote production overtaking the world, IP infrastructures and workflows are becoming increasingly prominent. Broadcast & CableSat (2021) states that broadcasters also start to implement more and more scalable, flexible, and reconfigurable solutions. This can be achieved using IP-based resolutions, although they need to find a solution that can be easily incorporated into already existing ecosystems.

3.4 Challenges and Obstacles

As with any technology, implementing and using AoIP requires thoughtful planning. Moreover, even with that, some challenges may arise. Proper network design is crucial to prevent issues such as latency, packet loss, and congestion (Vidal, 2019). Multimedia over networks, such as audio, are very susceptible to delay and its variations, such as jitter, that degrade its quality. However, multimedia over IP is far less sensitive to packet loss as long as it is not excessive. Small glitches can easily get lost in the media.

To ensure stable audio transmission, especially in real-time environments, dedicated VLANs, Quality of Service settings, and sufficient bandwidth must be provided. The whole network topology is also influenced by the number of devices in it and the number of endpoints. To fully utilize the power of AoIP, managed networks should be used with the ability to use network management systems.

Another key concern is security. Because AoIP systems are often integrated into enterprise networks, they must be protected against unauthorized access and interference. AETA Audio (2014) explains that this includes implementing firewalls, access control lists (ACLs), and encrypted transport methods, where applicable. These processes are enforced to protect the network from attacks such as denial of service (DoS) attacks, session spying, and other intrusions on the server. It is the job of system engineers and the IT department to set up these measures well enough so that they will protect the network but also clever enough so that they do not interfere with the flow of production.

3.5 Applications and Advantages

Although AoIP is still not a perfectly balanced technology, it is widely accepted in the world of broadcast and AV production. It replaces traditional audio routers and cabling in broadcast facilities with a more scalable and manageable solution. As the demand for better quality grows, better technology must adapt. Fiber optic cables can deliver audio with very little interference for up to 100 km (Kurose & Ross, 2017).

The shift into IP production does not have to happen all at once. Integration of a hybrid environment can take place by providing a mechanism to bridge disparate network segments, including connecting existing SDI infrastructure to the IP domain. This type of integration is managed by Software-Defined Networking (SDN) controllers which offer network security and access control. The centralized control plane ensures signal integrity and security by allowing dynamic and precise routing while strictly managing bandwidth and preventing unauthorized access to audio streams (Suard, 2020).

Lastly, broadcasting is not the only way to utilize AoIP. It can be used for live audio events, allowing for fewer cables and, thus, flexible stage layout and movement with fewer redundancies. It can also be used in education and business settings, as it can easily deliver announcements, conferencing, and multimedia presentations to many people.

4 MADI

A characteristic of modern media production workflow is a shift toward AoIP systems, which utilizes conventional network infrastructure to facilitate digital audio transport. However, the initial growth of AoIP led to protocol fragmentation, with prominent proprietary solutions such as Dante, Livewire+, Ravenna, and Q-LAN emerging. Although these systems offer high-quality multichannel audio transmission over IP networks, their differing architectures often result in limited native interoperability.

To ensure seamless integration and standardized connectivity within professional audio, the Audio Engineering Society (AES) has historically established key digital audio standards even before fragmentation. One of the first and most impactful was AES10, commonly known as the Multichannel Audio Digital Interface (MADI). It is a point-to-point, device-to-device digital audio interface created in 1991 (Eastty et al., 1991), and its last update was in 2020. Although not an IP-based protocol, MADI remains widely used in professional broadcast, live sound, and studio applications due to its robustness and simplicity.

4.1 Fundamentals

MADI is a transport interface that can transmit up to 64 channels of digital audio. The audio is uncompressed at a sampling rate of up to 192 kHz and 24 bits per sample. The slight inconvenience may be that the channels must use the same sampling rate. Unlike AoIP systems, MADI does not operate on network protocols or packets. Instead, it uses serial data transmission over a dedicated cable, simplifying setup and ensuring deterministic latency (Eastty et al., 1991).

There are two primary physical transport types for MADI, a coaxial cable with BNC connectors, or a fiber optic cable. These allow MADI to travel over cable for up to 3000 m, which makes it ideal for broadcasting or recording on location (d’Escriván, 2012).

4.2 Compatibility

Devices equipped with MADI have very low boot time, making it simple and effective. It is also plug-and-play, so no special management or configuration is needed. It is a widely supported standard that offers a quick and straightforward solution, utilizing light and well-working cables.

5 Dante

Audio systems have always required a significant number of point-to-point connections, resulting in a new physical single-purpose cable for each one. Dante was able to replace all those connections with a simple Ethernet cable and IP network that allows for connections and management. Digital Audio Network Through Ethernet (Dante) is a proprietary AoIP networking solution developed by Audinate. Since its launch in the mid-2000s, Dante has become the most widely used technology in professional digital audio. It delivers uncompressed, reliable multichannel audio with low latency between devices on the same network, no matter the distance between them.

5.1 Fundamentals

Dante is a series of technological solutions that manufacturers all over the world implement in their hardware and software. This can vary from simple microphones to broadcast consoles and servers. Dante splits digital audio into packets and sends it via UDP using RTP across an IP network. With audio, it also supports video transport. When a Dante device receives media packets, it reconstructs them back into a data stream.

For synchronization across devices, as stated by Audinate (2022) in their technological information for network administrators, Dante uses PTPv1 by default. One device is chosen as the Grandmaster, and the others follow it. A leader device with a 1Gbps connection can synchronize up to 250-300 devices. Dante works on a plug-and-play basis, which means that when a new device is connected to the network, Dante can automatically recognize it and integrate it. A requirement for connection is that the latency does not surpass 10 ms. Devices that are already part of the network can sometimes gain up to 40 ms of latency. Redundancy is also supported, primarily for applications that require high uptime, such as broadcast. Dante is, by default, set to unicast streaming, but it can be changed to multicast. Dante audio cannot be transmitted over WiFi due to high latency and very low reliability.

5.2 Compatibility

Although Dante is a closed proprietary networking protocol, it can offer compatibility with AES67 (Audinate, 2019). Because Dante predates AES67, its original implementation utilized PTP v1. Conversely, AES67 mandates the use of PTP v2, and the two versions are not directly interoperable. This difference presents a significant synchronization challenge when deploying both solutions on a single network, often necessitating the implementation of a dedicated PTP translation gateway or precise PTP domain management. Although Dante has subsequently integrated native support for AES67 and PTP v2 in its newer devices, this expanded functionality is not universally available throughout the product ecosystem.

Dante has developed a hardware solution for many other options, such as AVIO. It is a portable interface that converts Dante to AES3, XLR, USB, Bluetooth, and vice versa. When Dante devices are connected to the same network, they can be unmanaged. Users can choose where the data streams go, but devices operate independently without centralized management (Audinate, 2025). Audinate created a Dante Domain Manager to group the management of networks and

devices in one place. It is a software implementation that allows users to actively manage and control network flow. They can administer the network's security, create logical groups of devices other than physical infrastructure, control the functionality or errors in real-time, or share content across subnets.

6 AES67

The Audio Engineering Society introduced the AES67 standard in 2013 to enable basic interoperability between otherwise incompatible AoIP systems (Hildebrand, 2014a). It is one of the biggest interoperability standards issued by them, and it is nowadays an inseparable part of any AoIP network.

6.1 Fundamentals

“This standard defines an interoperability mode for transport of high-performance audio over networks based on the Internet Protocol suite.” (Hildebrand, 2014a, p. 1)

AES67 does not create any new standards. It is also not a whole media transport solution. It is simply a collection of guidelines that can be followed to achieve full interoperability between devices transmitting digital audio over IP. However, it is expected that different solutions will need some modifications to match them. It focuses on uncompressed, low-latency, high-capacity digital audio transport, as outlined by Hildebrand (2014a) in several of his works (Hildebrand, 2014b).

AES67 defines several key technical components for interoperability:

- Synchronization - Precise timing and synchronization are achieved using IEEE 1588-2008 Precision Time Protocol (PTPv2). This is crucial for aligning streams and maintaining clock stability across the network.
- Transport Protocol - Audio is transmitted using the Real-Time Protocol (RTP) over UDP/IP, ensuring low-latency packet delivery is suitable for professional audio use.
- Latency - AES67 specifies latency values between 1 ms and 10 ms depending on the implementation, with optional buffer values for jitter compensation.
- Audio Format - It mandates support for 48 kHz with recommended support for 44.1 kHz and 96 kHz sampling rate. Sending devices must support 16-bit or 24-bit depth, or both. The receivers must support both. If the device operates at 44.1 kHz, it must support 16-bit bit depth, and for 96 kHz, 24-bit support is required.

All of these elements are based on widely adopted IETF standards (eg, RFC 3550 for RTP), ensuring that AES67 aligns with broader networking practices and infrastructure.

6.1.1 Synchronization and Media Clocks

The synchronization between devices connected to the same network and their capability to share a proper common clock differentiates high-performance digital audio transmissions from those lower ones, such as voice-over-IP and Internet radio. A common clock guarantees the same sample rate and stream to all devices. This is critical for some network audio devices, such as mixing consoles.

AES67 uses the IEEE 1588-2008 Precision Time Protocol (PTPv2) for synchronization. IEEE-1588-2019 (2020), a newer version, explains that its epoch origin is 01. 01. 1970 00:00:00 UTC,

and it has been counting since the international atomic time. It also helps to establish a common wall clock across the network, as it has a Best Master Clock algorithm (BCMA). Another protocol defined is for distributing the correct time between the time receivers and the Grandmaster clock. It is not a requirement, but Grandmaster is usually synchronized to an external reference, like GPS.

To distribute the time information for the Grandmaster of the network to all other devices, PTP uses multicast SYNC messages. A delay mechanism is also used to achieve better accuracy of the data stream. With all this implemented, the achievable precision of the stream may reach dual digit nanoseconds.

In compliance with RFC 3550, a standard protocol for real-time connectivity, all RTP clocks with AES67, while representing the same sample counts, use a constant offset to their media clocks (Schulzrinne et al., 2003). Senders utilize these to sample, and receivers watch the media stream. AES67 instructs for support for 48 kHz sampling frequency. However, it is recommended that the devices also support 44.1 and 96 kHz.

6.1.2 Transport, Encoding, and Streaming

AES67 also addresses how to packetize and format audio into packets and how to transmit them across the network. The required and recommended sampling rates were already mentioned in the previous chapter. The packet time, the duration of the media contained in the packets and the sampling rate determine the number of samples per packet. The lower the packet time, the lower the latency, but also the higher the rate at which the device must receive the packets, which can overwhelm some devices. With longer packet times, there will be higher latency and increased buffering, which can be harsh on the devices' memory. The sender provides the packet time, although it may be negotiated. The packet time required for interoperability given by AES67 is 1 millisecond. Additional packet-time support is recommended.

The required buffering and its implementation can be network-specific, but under AES67, receivers must support a minimum buffer capacity of three times the packet time. The recommendation is at least twenty times the packet time or at least 20 ms.

Regarding channel counts, every receiver must support a stream with at least one to eight channels. Streams with more than one channel may be supported optionally. Senders must be able to offer at least one stream with eight channels or fewer.

Priorities must be established when streaming different data times, some real-time and some not. This is usually done by implementing Quality of Service (QoS) on the network. The requirement is that the network must support at least three classes and then tag them with the appropriate DSCP value (Hildebrand, 2014).

The Real-time Transport Protocol is used for media data transport, as defined in RFC 3550 / 3551 by Schulzrinne et al. (2003). Its packets are transmitted with UDP. AES67 also allows for the use of multicast messaging. They are used for synchronization or media streaming, as described in RFC 1112 by Deering (1989).

6.1.3 Stream Description and Connection Management

Engineers use stream descriptions to manage and install the network and its data streams. They contain information such as network addressing, encoding, media clocks, and more. Stream Description Protocol (SDP) represents important stream information for connection administration, as defined in RFC 4566 by Perkins et al. (2006).

Connection management establishes a data stream between a sender and one or more devices. Session Initiation Protocol (SIP) is utilized within AES67 (Schooler et al., 2002) for unicast. If a multicast is needed, a simple connection is established using the IGMP protocol without SIP. There is a serverless option for limited-scale installations. It is used to allow connection management between two devices, simply peer-to-peer. It is a requirement for AES67 devices to support serverless mode, as well as support for utilizing SIP in the network (Hildebrand, 2014).

6.2 Compatibility

Many popular AoIP platforms now support AES67 as a compatibility layer. For example, Audinate's Dante, AIMS' Ravenna, Axia's Livewire+, and QSC's Q-LAN can exchange audio with one another through AES67-compliant configurations. The broadcasting industry is much smaller than the IT industry. By utilizing and connecting it, such as with AES67, broadcasters are given much greater opportunities to expand further (McGinley, 2017). It allows for the integration of devices from different manufacturers while maintaining a simple unified network. Unfortunately, interoperability is usually limited only to audio streaming. Many features, such as device discovery or control commands, are still typically proprietary and may not function across other vendor platforms.

7 Use of AoIP in DSNG46

The application part of this bachelor's thesis focuses on the question of using digital audio transmission over IP in the real world of broadcast. More concretely, the DSNG46 (digital satellite news gathering unit) of the Czech Television and whether using this technology there simplifies the crew's work.

The author of this thesis is currently working in the above-mentioned broadcast company and was briefly established as a part of the group creating the project of a remodel of an outside broadcast van. Czech Television showed interest in using some of the AoIP technology before the remodel expanded, and further integration became a question for the company and the author. Czech Television, as well as many organizations worldwide, is slowly moving its production online. Collaboration tools, communication, and faster transport of more and better-quality data are required. The question that arose from this observation by the author was that if everything is moving online and getting better, then audio should, too. Thus, audio over IP will help simplify production, offer wider interoperability, and facilitate easier management and better quality of media transmission. This hypothesis inspired the work that followed.

In most of the literature on the topic of audio over IP and its interoperability, this technology is presented as a well-working structure with standards that offer easy implementation of devices. However, most of this information is provided by singular companies that present their own devices, which often include proprietary features. Even these minor differences can produce significant obstacles while creating a larger-scale ecosystem of devices.

Moving production online and offering broadcast crews greater flexibility may seem thoughtful, but it also comes with disadvantages. New technologies require more knowledge from the crew, sometimes even a new member. It also means that crews cannot exchange their vans without knowing every mechanism.

The following chapters will explain further how this new ecosystem changed the flow of production and the response from its users.

7.1 Current Situation

The main objective of this work is to establish whether the use of AoIP in the transmission of broadcasting cars in the production of Czech Television is useful. This company implemented AoIP in DSNG46.

Czech Television has a collection of OB vans of various sizes and for various purposes. DSNG cars are mainly used for live streaming, such as news reports, and, starting in 2025, even sports. The car is a van with all the needed technology inside. Most of the setup is already prepared, so preparation on the spot is supposed to be quick. A part of the broadcasting crew sits in the back of the car, managing the devices and data flows while broadcasting, and the director, who can be in a different location, communicates with them through an audio connection.

The van that is the main focus of this thesis was, two years ago, still classified as DSNG36. In 2023, Czech Television decided to upgrade the technology used in this car. After developing a plan for the public contract, it has become apparent that it will change a significant portion of the OB van, and there will be more new technology than the old one. Thus, a satellite certificate company gave the newly created DSNG a substitute license number 46.

This vehicle serves as the main outside broadcast van for the sports news department of Czech Television. Its function is to facilitate live transmissions from the location of various sports events. Due to the nature of these broadcasts, the OB unit must have the ability to integrate multiple external transmission feeds, often comprising both video and audio channels supplied by the event organizers. Furthermore, the vehicle must support the integration of commentator setup equipment to ensure the inclusion of live synchronous commentary during sports broadcast.

The interior of this van must accommodate a crew of two or three people, so the inside technology was chosen with space-saving in mind. It has influenced some of the decisions, mostly when it comes to digital mixers.

Figure 2

Inside of DSNG46 taken from the door for crew



Note The interior working space for the crew of DSNG46. Supplied by the author

The whole DSNG46 unit can be used as a standalone OB van or it can be used with a help car. To be able to function alone, a large amount of technology must be contained in it. This is achieved by resourceful planning and positioning of devices so that every bit of space is used to the maximum. As seen in Figure 3, the rack units are quite full, with spaces to allow airflow for cooling. The wall space is mostly used for displays, with an analog audio patch panel (top right corner in Figure 2) and a communications panel with a microphone.

Figure 3

Inside of DSNG46 working space



Note The interior working space for the crew of DSNG46 with a view of the racks. Supplied by the author

While broadcasting or recording with an OB van, stage boxes are often used. A stagebox is a central interface device used in live broadcast and live sound recording to group multiple audio (and sometimes video) signals from a stage or remote location and transmit them to a remote mixing console or control room via a single multi-core cable. It simplifies setup, minimizes cable clutter, and allows for a more organized workflow, replacing the need to run numerous individual cables from each source to the broadcast location. Stageboxes connect to the van from the outside, to be specific from the back of the car, as seen in Figure 4, to the panels in the lower right part of the vehicle. DSNG46 is equipped with two types of stageboxes, video and audio. All four of them in their preparation phase are shown in Figure 5.

Figure 4

The back side of DSNG46



Note The connection racks, storage space, and a look into internal cabling, seen from behind the car. Supplied by the author

Figure 5

Stageboxes of DSNG46



Note Four DSNG46 stageboxes situated within the Czech Television garage, undergoing connection validation and operational testing. Supplied by the author

7.2 Methods

To obtain the results of using a digital audio transmission over IP in DSNG46, the author used two qualitative methods: modeling and semi-structured interviews.

7.2.1 Modeling

To replicate and better understand the technological connections within the DSNG46 vehicle, several logical models were developed. These models categorize the systems and device interconnectivity based on their use of Audio over IP (AoIP) and related networking technologies.

Three primary groups were identified: PTP synchronization, Dante, and AES67. Key devices are represented within each categorical model. The significance of a device was determined by its role within the broadcasting workflow. Two additional models were included to illustrate the complex interconnectivity of the PTP and the audio production chain within the OB van. These representations are a simplification, as the actual connections are highly intricate and extensive. The final models presented in this thesis resulted from multiple consultations with subject matter experts responsible for managing the vehicle and its data flows.

The modeling served to create a simplified representation of the devices and connectivity used. It provides a comprehensive overview of the separate technology ecosystems in DSNG46 and helps to understand the inner workings and interoperability of the components.

7.2.2 Semi-structured Interviews

The author conducted four separate semi-structured interviews for this thesis. The goal was to get a better understanding of the choices made while creating the project of DSNG46, as well as to understand how it works with the OB van and AoIP. These interviews were conducted after the DSNG46 was in use for a few weeks so that the crew had some experience using it for real-life broadcasts.

7.2.2.1 Respondents

The four interviewed respondents participated in the development of the DSNG46 project and its planning. Each has a different field of expertise that is advantageous for this thesis.

The first respondent was Filip. He is a system engineer at Czech Television and has an IT consultant role. Since the new technology was implemented, Filip has been the network manager. Most of the projects he works on are an integration of new technology into the infrastructure of Czech Television broadcasts. His expertise is in networking and transmission control.

The second respondent was Vašek. He is an operator of DSNG vans for Czech Television. He has more than twenty years of experience working with various types of broadcasting cars. Vašek is also the head of the crew of DSNG46 when it is out of broadcasting. He was chosen as a respondent mainly to talk about his own experience with the new technology compared to the old one, which he had been used to for several years.

The third respondent was Jirka, who is a system audio engineer. He services most of the audio technology in Czech Television and has a broad awareness of what is used. He tested several of the devices that were later installed into DSNG46 and consulted on the use of new technologies that future users requested.

The fourth and last respondent was Tomáš. His position is that of a system engineer in the same department as the first respondent. He worked on several projects regarding OB vans and studio complexes. On the DSNG46, his role was that of a consultant.

7.2.2.2 Questions

The respondents were asked a series of open-ended questions that had been prepared in advance. However, not every question was asked from the list, as it was a semi-structured interview. The author customized the list of questions for each respondent to best suit their expertise. Each interviewee was asked to introduce themselves, their job title, their role in the DSNG46 project, and their experience with any other broadcasting vehicle.

System engineers were asked about the creation of the project and its planning, as well as how the choices about the technology used were made. They were also asked about the comparison of technologies in other broadcasting vehicles in Czech Television. After that, they discussed their opinion on audio over IP and its use in DSNG46. They explained the advantages and disadvantages of this technology. Lastly, the questions asked were about the changes that AoIP brought to the broadcast and its future use in the company.

A different set of questions was prepared for the respondents from the OB van's crew. This debate focused on their experience working with DSNG46 and the new technology installed compared to the older models. These inquiries focused on the new problems and benefits AoIP brought to their work environment. Lastly, the interview ended with their opinions on whether this technology should be used onward or if the production should stay in analog form.

7.3 Results

This chapter presents the outcomes of the applied portion of the thesis, which aimed to explore the practical implications of implementing Audio over IP (AoIP) in a Digital Satellite News Gathering (DSNG) environment. The hypothesis guiding this work was that AoIP would result in better audio quality, improved usability, and greater flexibility than traditional systems. To test this, five models were created showing device interconnection in a DSNG setup using different AoIP technologies: Dante, AES67, Precision Time Protocol (PTP), overall audio production chain, and comparison of PTPv1 and PTPv2. Additionally, qualitative data were collected through four expert interviews with professionals who work directly in television production.

7.3.1 Overview of the Models

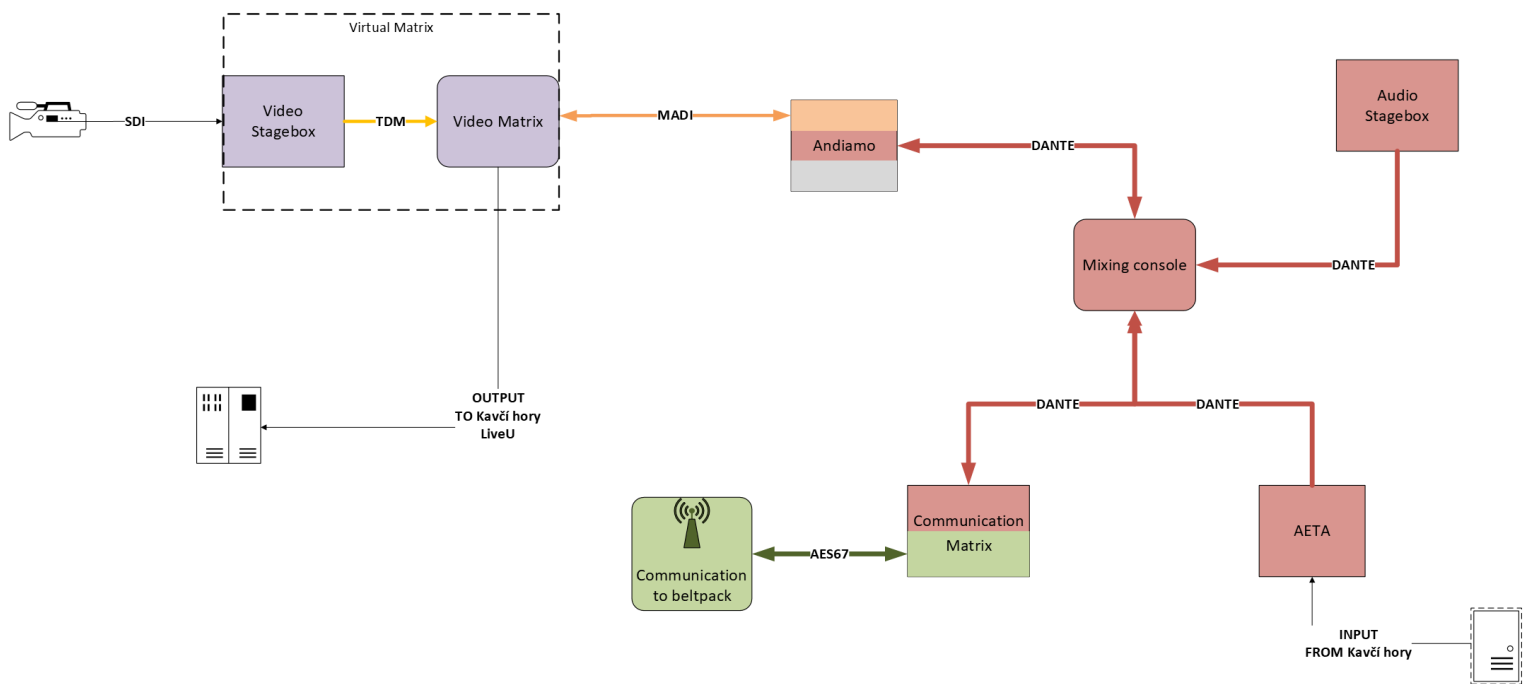
Each model illustrates the same core DSNG system - consisting of a digital mixing console SQ6, two main Arista 720xp switches that work as boundary clocks, intercom, and its components, connection to video servers, and more devices - but implemented using a different underlying audio transport technology. Some of these models utilize overlapping sets of devices. This allows

for interoperability between technological solutions. Some of these connections are pre-built into the device, but some must be achieved by adding converters to the production chain.

Model 1: Audio Production Chain

Figure 6

Audio Production chain in DSNG46



Note Schematics of the audio production chain in DSNG46. Own work

To accurately illustrate signal flow and interconnectivity between all components of the DSNG46 audio chain, a simplified graphical model was developed. This model represents all devices involved in audio processing during production from the OB van, as well as the specific interfaces employed for signal transmission. Table 2 provides additional information, detailing the devices' names, types, and a concise overview of their function.

Four primary input sources are available: SDI feeds from cameras or other transmissions to the video stagebox, wireless beltack communication links with remote crew members, a transmission channel to the audio stagebox, or an external feed originating from the main Czech Television facility. Transmission output to broadcast is singularly channeled through the main Video Matrix.

The mixing console is centrally located within the audio chain, as seen in Figure 6, and is configured for Dante operation in AES67 compatible mode. This configuration allows the Audio Engineer to consolidate all input sources and execute the requisite mixdown. A key feature is the integration of communication channels with the mixing console, which enables the Audio Engineer to output mixed audio to communication circuits without routing the signal to the main broadcast feed.

Table 2*Overview of the devices used in DSNG46*

<i>Name of the device</i>	<i>Type</i>	<i>Brief overview</i>
Video stagebox	Riedel MicroN + Direct Out Andiamo 3 + Arista 720 DP - 24S	A small stagebox unit with added compatibility rack units. Equipped with SDI output and input, Analog Mic/Line Inputs, Analog Line Output, AES3 Digital Audio Ports, MADI optical ports, Ethernet ports.
Spine Switch	Arista 720xp - 48YP	High-end network switch.
Mixing console	Allen & Heath SQ6	Small size mixing console with touchscreen, 24+1 faders, Dante. Input/Output through added audio stagebox: Mic Preamp Inputs, Line Outs, 48 kHz and 96 kHz compatibility, Dante.
Andiamo	Andiamo 3	High-end Analog-Digital and Digital-Analog converter, equipped with Analog audio, Dante, MADI.
AETA	Multiscoop 2	Modular multi-channel Audio over IP codec for connection with Kavčí hory. Can be AES67/Dante enabled (used in Dante form), mono/stereo audio channels.
Communication Matrix	Riedel Artist 1024	Communications platform compatible with other ČT studios. Compatible with Dante, AES67, MADI.
Video Matrix	Riedel MicroN UHD	Small size audio and video matrix. Includes MADI input, SDI input and output, 16 Channel Audio Embedder and Deembedder, and Built-in Sample Rate converter.
Audio Stagebox	Dante with DT 168 I/O	A portable I/O extender with 48 kHz – 96 kHz compatibility, Dante, XLR outputs, line level preamps. Is usually used as an extension for remote I/O for SQ6.

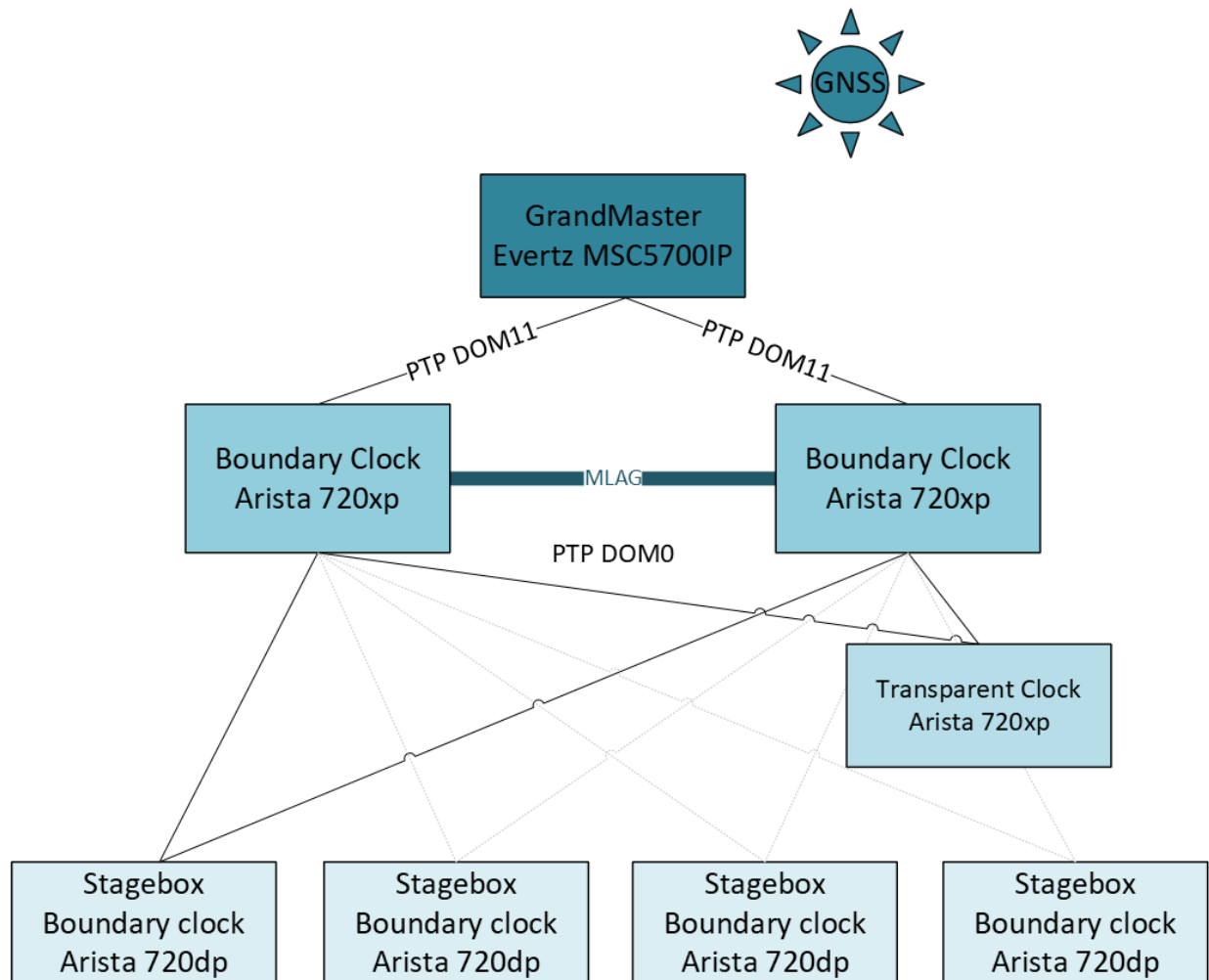
Note A table with a brief overview of the devices used in the audio production chain of DSNG46. These devices were chosen as part of a public contract by the contractor that offered the lowest price. Data for Spine Switch are from *Arista 720XP Series*, by Arista, 2023, (<https://www.arista.com/en/products/720xp-series>). Copyright 2023 by Arista. Data for Mixing console are from *SQ-6 • Allen & Heath*, by Allen & Heath Limited, 2025, (<https://www.allen-heath.com/hardware/sq/sq-6>). Copyright 2025 by Allen & Heath Limited. Andiamo data are from *ANDIAMO - High-End AD/DA Converter | DirectOut*, by DirectOut, 2025, (<https://www.directout.eu/product/andiamo>). Copyright 2025 by DirectOut. Data for AETA are from *Modular multi-channel codec with IP, ISDN and AES67*, by AETA Audio Systems, 2022, (<https://www.aeta->

[audio.com/downloads/doc/product_data/MultiScoop_ENG.pdf](https://www.riedel.net/fileadmin/user_upload/800-downloads/02-Brochures/EN/ARTIST_-_The_Intercom_Brochure.pdf)). Copyright 2022 by AETA Audio Systems. Data for Communication Matrix are from *ARTIST THE INTERCOM* by Riedel, 2024, (https://www.riedel.net/fileadmin/user_upload/800-downloads/02-Brochures/EN/ARTIST_-_The_Intercom_Brochure.pdf). Copyright 2024 by Riedel. Data for Video Matrix and Video Stagebox are from *MediorNet MicroN MediorNet MicroN The 80G Media Distribution Network*, by Riedel, 2015, (https://www.riedel.net/fileadmin/user_upload/11-products_neu/media-networks/micron/Hardware/06-210DB00EG-A10_Datasheet_MN-MicroN-2.pdf). Copyright 2015 by Riedel Data for Audio Stagebox are from *DT168 • Allen & Heath*, by Allen & Heath Limited, 2025 (<https://www.allen-heath.com/hardware/everything-i-o/dt168/>). Copyright 2025 by Allen & Heath Limited.

Model 2: PTP Synchronization Architecture

Figure 7

PTP synchronization in DSNG46



Note Schematics of PTP synchronization in switches in DSNG46 and their interconnections.
Own work

The first created model was that of PTP synchronization on the network, as this was the most complicated at the time of creating DSNG46 and was needed the most. This model focuses less on a specific AoIP protocol and more on evaluating the timing infrastructure. An underlying fact that is not displayed in the model is that the technologies in the van that use PTP do not use the same version. Thus, the implementation of this technology required complex network management to make these two versions compatible so that every network device can be adequately synchronized with its given protocol.

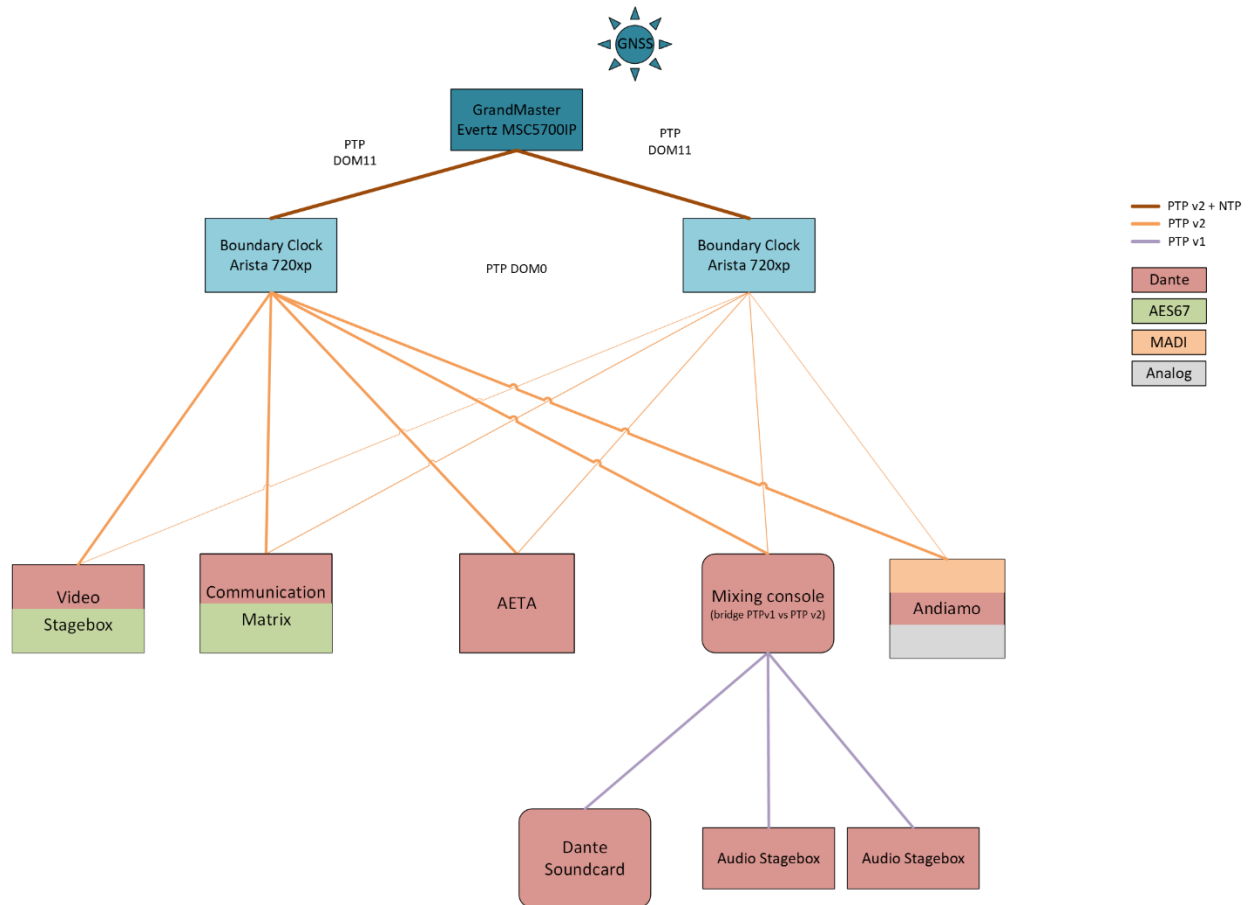
The network connection for synchronization is split into separate domains so as not to overwhelm the leading devices, as seen in Figure 7. The main world clock generator, an Evertz Grand Master, gives out the primary impulses and leads other devices beneath it. It is also connected to a GNSS antenna located on the van roof for additional synchronization. Two Arista switches are connected to the Grand Master. These switches are boundary clocks. This means that only they can communicate with the device above them in the topology, and the devices beneath communicate with them. The further communication is then again split into separate domains. The OB van was required to have four separate stage boxes that could be joined when needed to broadcast. This was why a separate switch was added to every stage box, which also acts as a boundary clock, as explained by the first respondent.

This model is particularly relevant for displaying synchronization behavior in live DSNG conditions. A significant amount of digital audio transport relies on PTP to avoid timing drift, and the topology includes several boundary clocks in groups to maintain synchronization across network segments. In practice, this would be part of a larger system using AES67 or SMPTE ST 2110, but here it serves to isolate the clocking aspect of the network.

Model 3: PTP v1 and PTP v2 Split Architecture

Figure 8

PTP v1 and PTP v2 in DSNG46



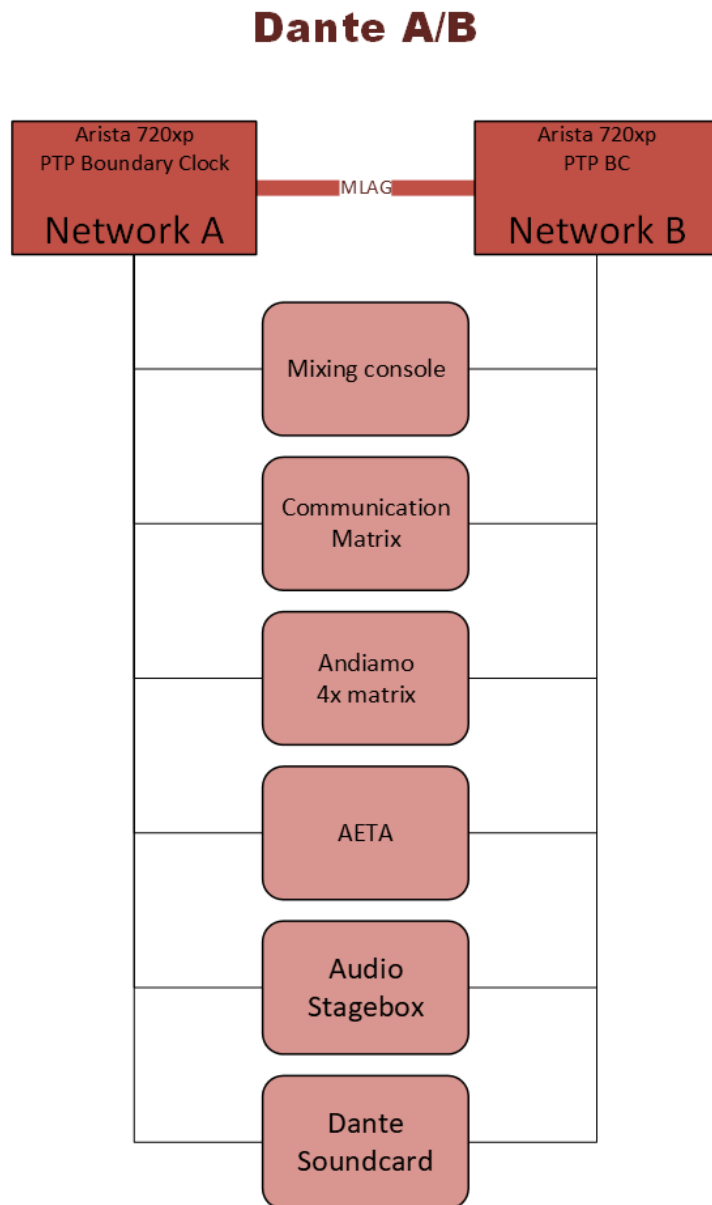
Note A topological analysis of the DSNG46 devices is presented, differentiating the PTP synchronization architecture based on the required PTP v1 and PTP v2 compatibility. Own work.

To further analyze the PTP architecture, a model was developed that illustrates the difference between PTP v1 and PTP v2 connections. This model depicts the logical structure required for synchronization interoperability across the devices. Since not all devices within DSNG46 support PTP v2, a PTP translation mechanism, a bridge, was required. In this configuration, the mixing console acts as a PTP translation gateway, receiving the PTP v2 signal from an Arista boundary clock switch. This bridge then translates the PTP v2 signal into a PTP v1 signal for distribution to non-PTP v2-compatible devices. Figure 8 illustrates that a subset of devices utilizes NTP concurrently with PTP. This configuration is limited exclusively to the main spine switches; all remaining devices rely solely on PTP for synchronization.

Model 4: Dante-Based Architecture

Figure 9

Dante-based network in DSNG46



Note Schematics of Dante-compatible systems in DSNG46 and their interconnections. Own work

This model uses Audinate's Dante protocol to interconnect all devices via a managed Ethernet switch. Dante supports plug-and-play configuration, device discovery, and channel naming. The model demonstrates significantly reduced cabling complexity: all audio and control data are

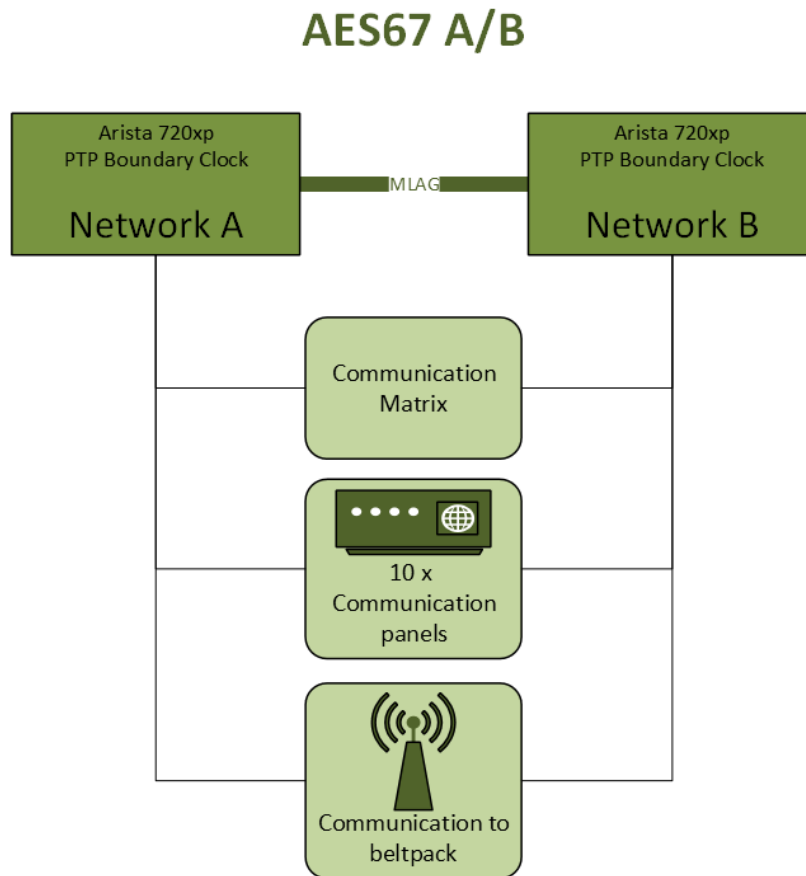
carried over standard CAT cables through a single network infrastructure. Devices, including mixer and intercom, are configured and routed using Dante Controller software or a device's internal matrix, allowing operators to change routing without physically reconnecting cables.

Dante is also connected to ANDIAMO, a high-end analog-to-digital and digital-to-analog audio converter (DirectOut, 2025), and to AETA, a wireless high-speed communication solution (AETA Audio Systems, 2025), as seen in Figure 9. These technologies are used to add different types of audio to the network, an analog option, and a codec over IP, enabling communication and connection with remote parts of production, as well as their contribution.

Model 5: AES67-Compatible Systems

Figure 11

AES67-compatible systems in DSNG46



Note Schematics of AES67 compatible systems in DSNG46 and their interconnections. Own work

This model was based on AES67, an open interoperability standard supported by many professional AoIP devices. In contrast to Dante's proprietary system, AES67 focuses on enabling devices of different manufacturers to exchange audio streams without needing an internal card.

Initially, the DSNG46 network was supposed to be connected to AES67 as its leading technology. Unfortunately, this was not possible. Now, it is primarily used for interconnecting the needed devices that do not support Dante and cannot be connected by analog cables with the rest of the van. These devices are mainly used for communication within on-site production or with remote parts of the crew, as indicated in Figure 10.

7.3.2 Interview Results

Four semi-structured interviews were conducted with professionals currently working in Czech Television. Each interview explored the participants' experience with DSNG cars, audio routing, and general impressions of AoIP technologies in live mobile production.

Scalability and Flexibility

The respondents emphasized the scalability and flexibility of AoIP systems several times. They highlighted the ability to use one Ethernet cable for up to 64 audio channels, which is a significant change from using audio cables. As engineers mentioned in interviews, the demands for broadcast productions are increasing. The company is pressed to provide more channels and better sound quality.

The difference mentioned several times was the size and price of the devices, as well as their connections. AoIP-compatible devices and Ethernet cables are much cheaper than proprietary analog audio technology. Cables are also more accessible, easier to install, and take up less space. This is very important, as DSNG units are small in size, even though their requirements, in some cases, almost match those for large broadcast trucks. As DSNG46 is only 7 meters long, the space for working with audio is around 1 meter, so the devices had to be very compact, as mentioned by the last respondent. The engineers also mentioned an advantage, which is that most systems like Dante are plug-and-play, meaning that they can connect to new devices without significant changes to the network.

A disadvantage that arises from using IP networks to deliver audio is the connection to other AoIP systems. Although it may seem simple, as the need for cables is reduced and compatibility is high, the networks are not entirely compatible if two systems use similar IP addresses for their links. These issues can have a profound impact on the broadcast as it makes the connection impossible. This had already proven to be an issue with a different OB van of Czech Television when there was a collision of IP addresses in the car and the stadium, which disrupted the whole network, as the fourth interviewee revealed.

Management

A fact mentioned by all respondents was that signal management requires a more profound knowledge of networking; therefore, it is harder for the crew to operate on their own. The benefit is that there can be pre-defined settings that they can bring up and use. A network engineer is able to prepare those before the broadcasts, so they are easily accessible and straightforward to use. Unfortunately, this is prone to human error, as even a slight change in the network can disrupt the connection matrix.

A crew member also mentioned in the interview that locating a mistake in the links has proven more challenging for them. With an analog connection, they could physically see a problem with a cable. With networks, troubleshooting is more inconvenient, as the problems are usually in the software. He also added that the crew cannot find and solve the connection troubles themselves, and they need the help of a more knowledgeable person in the network. An audio engineer

agreed with this statement, as he felt that the system was not well assembled and that there were many places where mistakes are bound to occur.

Sound Quality

None of the respondents noticed any significant improvements in audio quality. It is worth mentioning that, as referenced by the first interviewee, the number of analog/digital conversions was significantly lowered. The whole system inside the DSNG unit uses uncompressed audio, which should yield better sound quality. Unfortunately, as the sound is transmitted to the primary production in Prague, compression must happen over the networks. Furthermore, when the data are streamed for intraterrestrial or internet broadcast, it uses lossy compression. Because of all these factors, the viewer will probably not recognize the quality change, as agreed upon by the respondents.

7.4 Analysis and Discussion

This chapter interprets the findings presented in the previous section considering the research hypothesis and the existing literature. It evaluates the strengths and limitations of each modeled system and its technology and reflects the interview responses gathered from television professionals. The aim is to contextualize the practical value of AoIP in mobile broadcast production, consider its broader implications, and identify areas for future improvement or investigation.

7.4.1 Overview of Key Findings

The results from the applied part of this thesis, including five models (PTP synchronization, PTP v1 and PTP v2 overview, Dante, AES67, audio production chain) and four expert interviews, mainly support the hypothesis that AoIP brings meaningful benefits to DSNG workflows. It also shows that the answer to the question of a more flexible broadcast system is not binary and that not everything is drastically improved by using IP networks. Specifically, AoIP systems were found to improve system scalability and flexibility and to be a cheaper and less bulky option compared to traditional analog or digital point-to-point systems. However, they also proved more complicated to manage and prone to human error. Unfortunately, there was no noticeable change in sound output quality. The findings highlight the importance of well-organized synchronization and signal management and the role of interoperability in determining real-world success with AoIP technologies.

7.4.2 Use of Models

Each model provides insights into how the AoIP technologies and devices that use them are interconnected inside the DSNG46.

Audio production chain

The audio production chain model underscores the necessity of a strategically engineered system in which the mixing console maintains signal input and output capabilities at all potential destinations. Furthermore, the model illustrates how various technologies often

operate as isolated domains or "islands", interconnected exclusively by a limited number of dedicated translation gateways or bridge devices. This topological complexity emphasizes the requirement for several connection protocols and interfaces to successfully facilitate a single audio transmission from the initial input to the final broadcast output.

PTP

The PTP-focused model underscored the importance of accurate clocking. While not an AoIP protocol, PTP is fundamental for all real-time media streams, and its successful deployment is critical to preventing audio artifacts such as clicks and phasing issues (Orme, 2023). This model was used for the planning body to better understand how the devices will connect and what the placement of master and boundary clocks in the topology is.

PTP v1 vs PTP v2

To further illustrate the interoperability and interconnectivity between various timing technologies, a model detailing the delineation between the PTP v1 and PTP v2 topologies was developed for the thesis. This model required multiple revisions to accurately reflect modifications to the vehicle's internal configuration resulting from operational use. The final version demonstrates how PTP v2 is distributed robustly and redundantly across all compatible devices. On the contrary, the PTP v1 signal must be channeled through the mixing console, which functions as the only PTP translation gateway. This dependence on a single device introduces a single point of failure into the synchronization architecture for all PTP v1 clients, making these connections vulnerable to disruption.

Dante

The Dante-based model emerged as the most practical and was used for actual broadcast data streams. Dante is still complicated for the crew members to understand, so this simple overview helps them navigate which devices use this technology. Unfortunately, Dante proved less interoperable than the literature, such as Audinate (2019), would suggest. The technology had to be set up in a special AES-compatible mode, and some devices could not be connected altogether.

AES67

The AES67 model demonstrated the long-term value of open standards, offering vendor-agnostic flexibility. Its use of communication greatly increases the possibilities of remote cooperation between crew members.

Joint work

Together, these models illustrate that while AoIP offers clear advantages, its success is contingent on proper network design, device compatibility, and synchronized timing. Thanks to their simplicity, they can also be used to introduce new colleagues to the inner workings of

DSNG46 within Czech Television. On the other hand, they cannot be used for professional use or exact network management, as they are far too elementary. However, that would defeat the purpose for which they were created in the first place. These models also do not showcase the interoperability between the technologies used. The reason was that the models did not intersect enough to be valuable.

7.4.3 Interviews

The interview responses reinforced the benefits observed in the models. All professionals interviewed highlighted the advantages of AoIP flexibility, scalability, and audio integrity. The disadvantages mentioned were complicated management and interoperability with other systems.

Limitations of the Devices Picked

A part of the interviews with members of the planning body was how they were limited in choosing the devices in DSNG46. The creation of DSNG46 was a public contract, so engineers could not choose the devices directly. They were working with a distributor. The most considerable influence was on the price of the products and then on their size. Sometimes, the cheaper option was to buy two converters instead of a device that would support both AoIP solutions, such as the mixing console. SQ6, which is the device that is installed as a mixing console, does not support AES67, only Dante in AES-compatible mode. Unfortunately, this option was too big to fit into the OB van. If the budget had been higher, devices with better compatibility would have been installed, and the production flow would have been improved. The space in the van is very limited, as mentioned by all the respondents.

Connections

As the demand for television broadcasts increases, requiring greater channel counts and scale, Audio over IP provides several solutions to meet these capacity requirements. As one CAT cable can carry tens or even hundreds of audio channels, the number of cables and the space saved is a considerable benefit, as mentioned by every respondent. As DSNG cars are pretty small, saving space is important. Even devices that use AoIP are smaller than analog audio, as they need fewer input and output options. However, once all ports on a device or switch are full, adding a new one would require a significant change in the topology and configuration.

A benefit that was mentioned several times was that the length of cables that can be used to connect devices has increased dramatically since the use of AoIP. While this is a great advantage, it may not be as usable in broadcast as it may seem. The OB van must travel to the location and connect to the local network, in most cases. Thus, a cable connection of hundreds of meters is not needed.

A problem that has already happened in the broadcast of Czech Television and a disadvantage of AoIP mentioned on several occasions during interviews is that connecting two separate topologies is troublesome. Most devices utilizing audio over IP are not produced with the thought of two interconnecting ecosystems. As mentioned by the audio engineer questioned,

these technologies, as made nowadays, are used for single-user production, like theaters or studios. This is a good point that vendors should integrate into their technology if they want to continue to thrive in interoperability. There are separate devices that can help avoid IP address collision and overall interconnection between systems, but they add another point where problems can emerge.

Demands on the crew

The crew member mentioned an interesting distinction between analog and digital audio systems. In analog connections, one manages the system device by device. However, AoIP must be managed as a whole. It gives an excellent overview of the whole process but makes it more complicated. Czech Television did not offer any new certifications or courses on how to manage the systems to the crew. Engineers only explained this when they created the setup. These new demands on the crew and the lack of support from the company are an oversight. The crew should be given thorough documentation of the technology used or a new member who is proficient with these devices and connections. All respondents mentioned that they believe that as broadcasting evolves, it will require more experts, especially in IT.

Management of Audio

The management of signal routing has proven to be difficult for the crew. The engineers have created premade setups that they can use, although it is just a matter of time until one of the presets breaks, and nobody currently in the van will be able to fix it. Even a change in cable can cause tens of channels not to connect correctly. This is a significant flaw in the interconnectivity of AoIP systems.

As a respondent said, AoIP production gives crew members much freedom, which may not always be beneficial. He also mentioned that an option would be to limit the possibilities of what the crew can change. As the software plays a significant role in routing, it may be possible to lock down some permanent routes.

Reliability is the most important aspect of a setup for the production crew. When they set up to broadcast, they need everything to work in minutes, as the first two respondents explained. This may become a problem in future production. Having an engineer check the preset between every type of transmission would be beneficial.

Communication

Czech Television implemented IP-enabled audio several years ago for the purpose of remote communication between parts of the production crew. In the broadcast of DSN46 in May 2025, almost 30 intercom links were used, as recalled by the first interviewee. This would require a high amount of point-to-point cable connections if done with analog audio, which would not fit into the small space of the van.

The AoIP connection also enabled the use of a mixing console with an intercom. This means that the audio mixer could reroute the communication through the console, add any necessary changes, and then send it to a reporter's in-earpiece.

7.4.4 Limitations of the Work

Although the findings of this thesis are promising, several limitations must be acknowledged.

The models created are schematic and may not capture the full range of complexity found in larger, multilocation DSNG operations. Although they offer a simple overview, they are custom-made for the specific use of DSNG46 in Czech Television. They are not suitable for the demonstration of connections with other broadcasting units.

Some factors that have limited the results of interviews and their analysis include the sample size. The interview data, while valuable, are limited to four participants. A broader sample might reveal additional insights or counterpoints. The size of the planning group was limited, but not all members were asked to participate in the interviews. In addition, not every member of the DSNG crew was questioned, as some of them did not feel qualified enough to answer.

A significant circumstance that must be considered is that the interviews were conducted two weeks after the first broadcast was streamed from DSNG46. This was enough time to allow the crew to get accustomed to the technologies, but not enough to fully understand them. Even a respondent from the crew mentioned during the interview that he was curious about how he would respond in a year. Unfortunately, the circumstances of this thesis did not allow for later interviews to occur.

The choice of devices also limited the DSNG46 and its technology. As the project was a public contract, many aspects, such as price and department requests, influenced the choice of devices. These decisions are described in the chapter above.

These limitations mean that, while the results are indicative, they should be interpreted with caution and viewed as a foundation for further, possibly larger-scale investigation.

7.4.5 Implications for Practice

Despite these limitations, the study provides a helpful reference for DSNG operators, broadcast engineers, and system integrators. The findings suggest that AoIP can simplify mobile production setups, improve audio performance, and allow easier reconfiguration during events if done right.

It also highlights the importance of selecting an appropriate AoIP solution depending on project needs: Dante for fast setup and closed environments, AES67 for long-term interoperability, and careful planning of PTP clocking across all models.

The fact that all respondents agreed on is that analog audio still has its place in production. Whether for its reliability or simplicity, it cannot be completely replaced yet. Its use will decrease as video over IP is more used.

A welcome change that would be beneficial but has not yet been observed is the change in broadcast audio quality. Both the input and output are significantly compressed. However, this would require a whole system change in the production of Czech Television, for which the company is not yet ready.

A list of recommendations was prepared for Czech Television to make working with new technologies, such as AoIP easier:

- Further certification. It is essential to implement a strategy to provide professional development and certification to crew members tasked with operating the new technology. Currently, the company has not established or advised personnel on the training protocols or resources necessary to achieve operational proficiency with the new equipment. This represents a significant deficiency in the deployment and learning methodology, directly impacting the ability of staff to function professionally with the system. It should be noted that some equipment manufacturers offer complimentary certification programs that could mitigate training costs; however, this opportunity remains unutilized by the crew, as they are unaware it exists.
- Planning with the crew. To mitigate future confusion, part of the crew members should be integrated into the planning committee to allow for their input and professional perspectives. Although personnel are occasionally consulted about suggested technology, their current level of involvement in the strategic planning process is insufficient. Consequently, upon completion of the project, they are often surprised by the final picks of the devices.
- IT education. The selection of the DSNG46 vehicle, which operates exclusively on a network-based infrastructure, for personnel lacking a foundational IT background, presents a significant operational challenge. To efficiently utilize all of the capabilities of the vehicle, existing crew must receive specialized training in network administration and management protocols, or the team must be expanded by a dedicated network engineer. Currently, all network management relies on one engineer located at Kavčí hory, but this situation is an unsustainable model that introduces substantial vulnerability to the production workflow.
- Uncompressed transmission to Kavčí Hory. AoIP has the ability to provide high-quality low-latency audio, but this is lost when every audio transmitted to the main facility is heavily compressed. To provide better sound to the audience, new transmission technology should be implemented.

7.4.6 Suggestions for Future Work

Future research could expand this study by conducting performance benchmarking between AoIP protocols in live broadcast conditions. This study focused on qualitative data and how DSNG46 and its technologies work. The sound and network's exact speed and other qualities have not been measured.

A guide could be created to help the OB van crew troubleshoot the AoIP systems. Beneficial research would be to conduct interviews with members of the DSNG46 crew after one year of

using it. It would bring new experiences and reveal flaws in the system that are not found shortly after the deployment.

8 Conclusions

This thesis set out to examine the role and potential of AoIP technologies within the specialized context of DSNG. The core hypothesis proposed that AoIP could significantly enhance DSNG production by delivering better audio quality, simplifying setup and routing, and offering greater system flexibility.

Through the construction of five technical models – audio production chain, PTP topology, PTP v1 and PTP v2 architecture, Dante, and AES67 - and four semi-structured interviews with engineers and operators working directly with DSNG systems, the study demonstrated that AoIP indeed presents tangible advantages, but also a few shortcomings. Each model showed simplified technological solutions and their connections in DSNG46.

Expert interviews supported the hypothesis of the benefits of audio over IP for television broadcast, but also highlighted several shortcomings of the technology. Respondents consistently highlighted improvements in scalability, reductions in cabling complexity, and an overall increase in operational agility. However, they also acknowledged challenges, particularly in configuring interoperable systems, and difficulties in network management. This emphasizes that proper planning, training, and understanding of network principles are crucial to success.

AoIP delivered on its promise of operational agility, achieving substantial improvements in system scalability and reducibility of cabling complexity. This efficiency, particularly the capacity for a single CAT cable to carry hundreds of channels, is essential for the space-constrained DSNG environment. The hypothesis of simplified workflow was only partially supported. The study highlighted persistent challenges, specifically the complexity of configuring interoperable systems and the difficulty of network management for non-IT-specialist crew members. This proved to be the biggest challenge facing the crew and the lack of knowledge of the network needs to be addressed.

The results underscore that the success of AoIP is dependent on the right planning and technical expertise. While Dante proved the most practical protocol for current broadcast data streams, the need for a dedicated PTP translation mechanism emphasizes the ongoing synchronization challenges when integrating heterogeneous timing technologies.

In conclusion, the shift to AoIP in DSNG operations represents a fundamental change in workflow philosophy, prioritizing adaptability, scalability, and efficiency over traditional point-to-point reliability. Future research, including real-world performance testing and additional interviews after prolonged use, is essential to refine best practices and fully unlock the system's potential.

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Appendices

Appendix A: Electronic appendices

This thesis includes four electronic appendices.

1. Recording of interview with the first respondent : *Interview_CM_1.wav*
2. Recording of interview with the second respondent : *Interview_CM_2.wav*
3. Recording of interview with the third respondent : *Interview_CM_3.wav*
4. Recording of interview with the fourth respondent : *Interview_CM_4.wav*